

Using Encryption to Enhance Confidentiality and Integrity (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 05

Student:
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Time on Task:
5 hours, 36 minutes

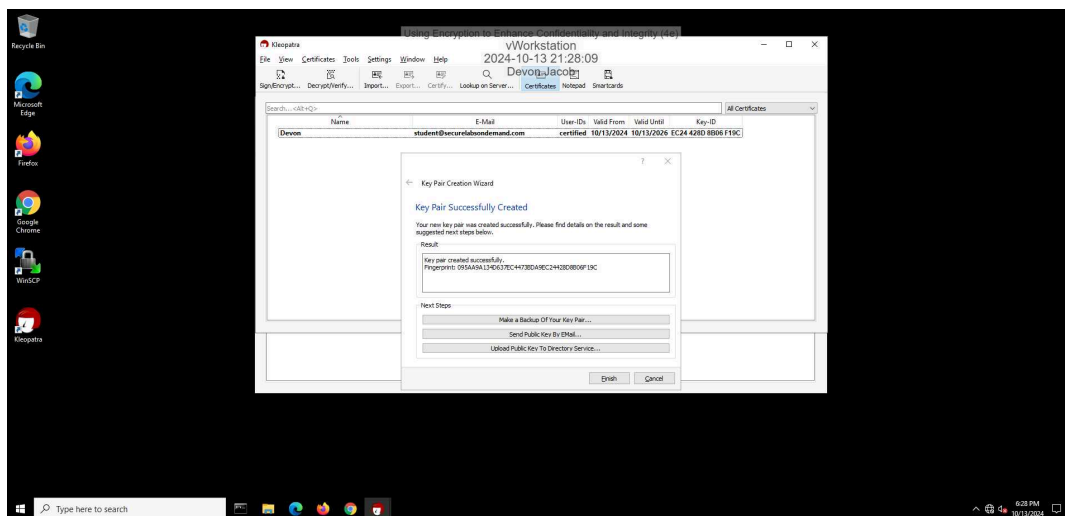
Progress:
100%

Report Generated: Sunday, October 13, 2024 at 11:12 PM

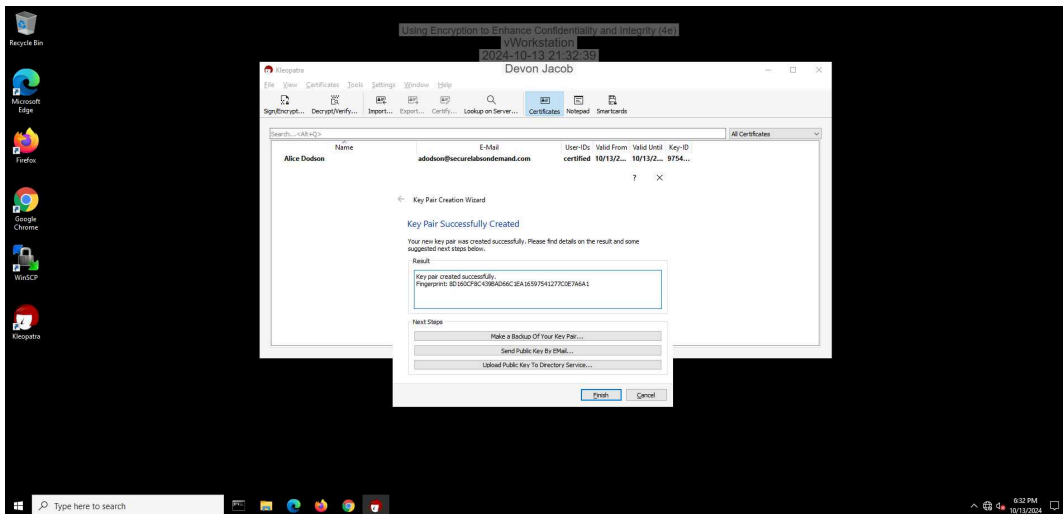
Section 1: Hands-On Demonstration

Part 1: Create and Exchange Asymmetric Encryption Keys

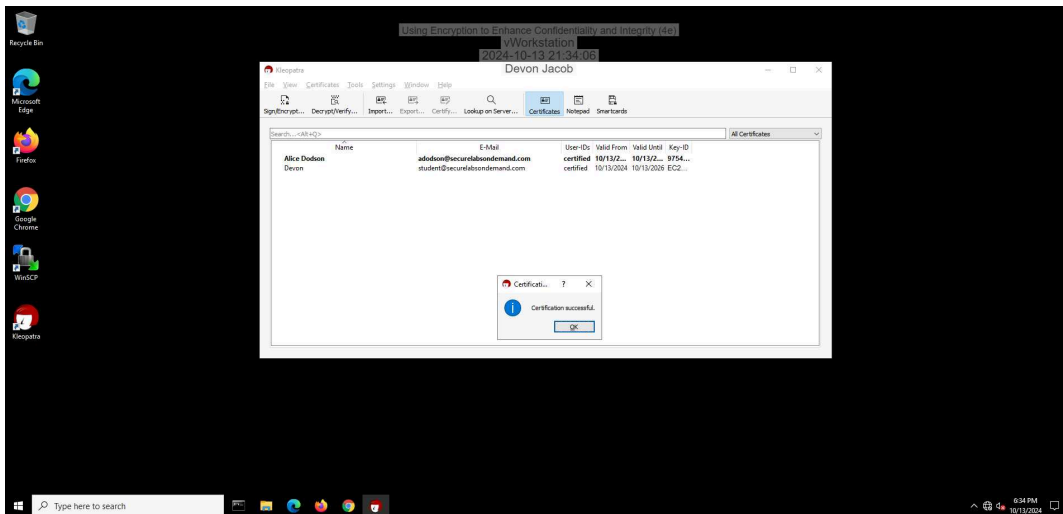
9. Make a screen capture showing the **fingerprint** for your key pair.



22. Make a screen capture showing the fingerprint for Alice's key pair.



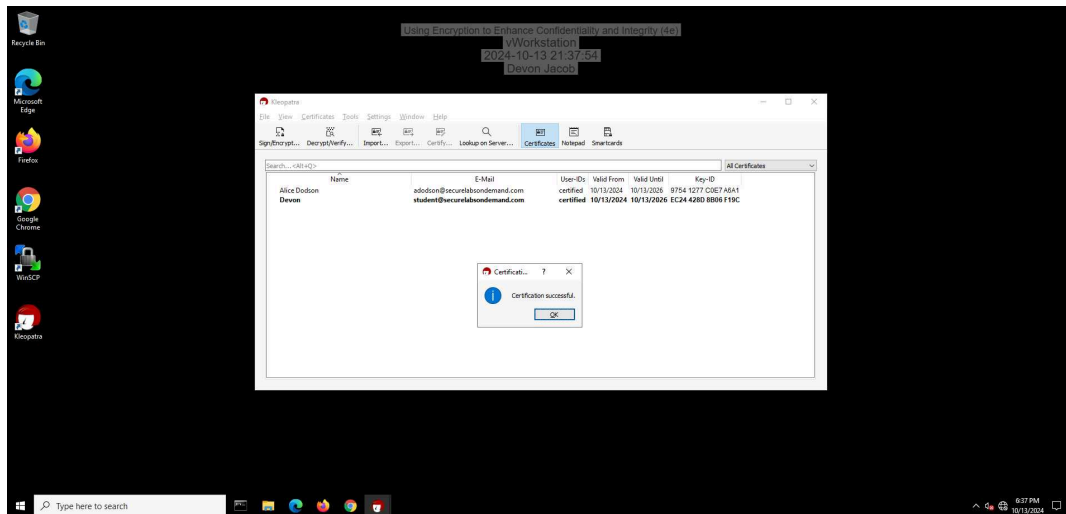
30. Make a screen capture showing your public key in Alice's certificate cache.



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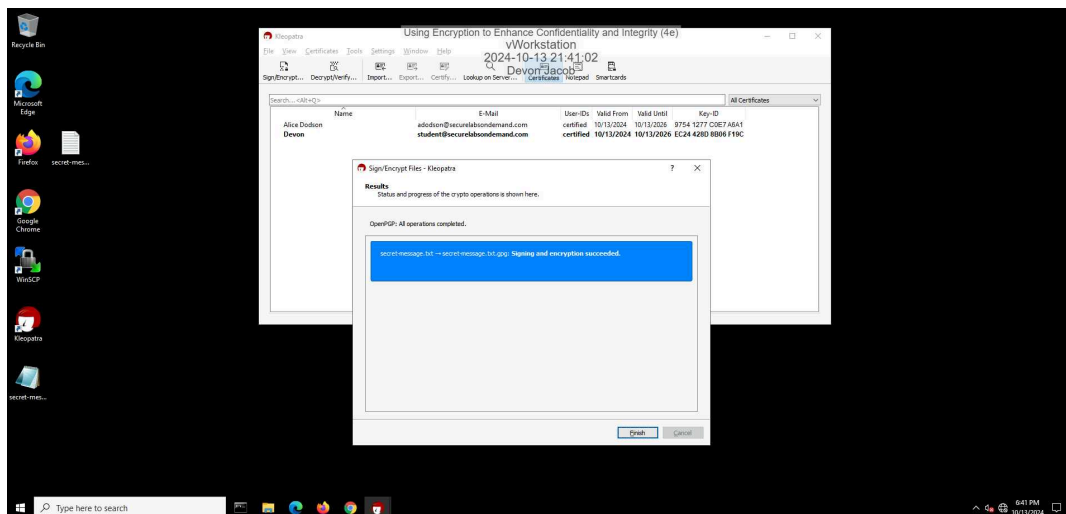
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35. Make a screen capture showing Alice's public key in your certificate cache.



Part 2: Encrypt a File Using Asymmetric Encryption

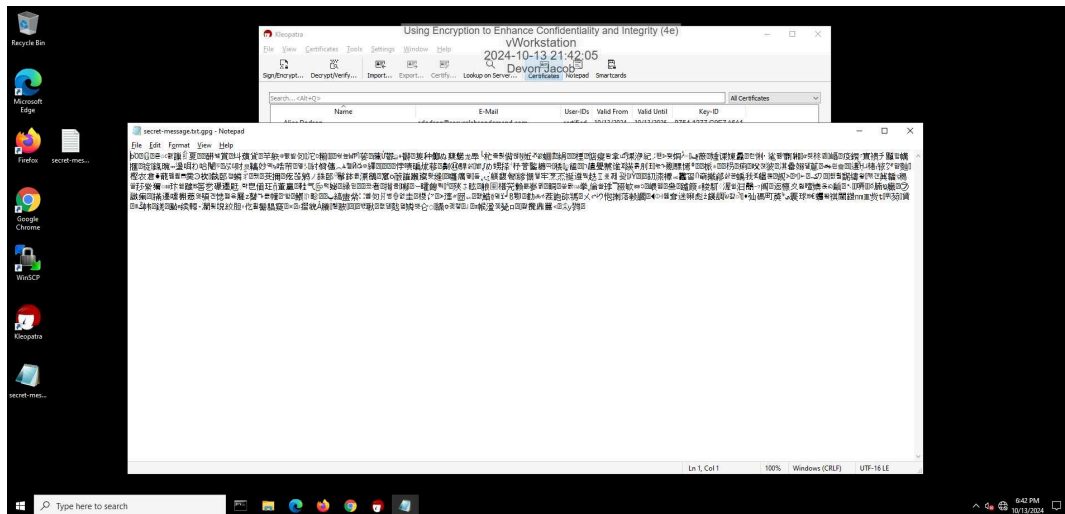
9. Make a screen capture showing the successful signing and encryption message.



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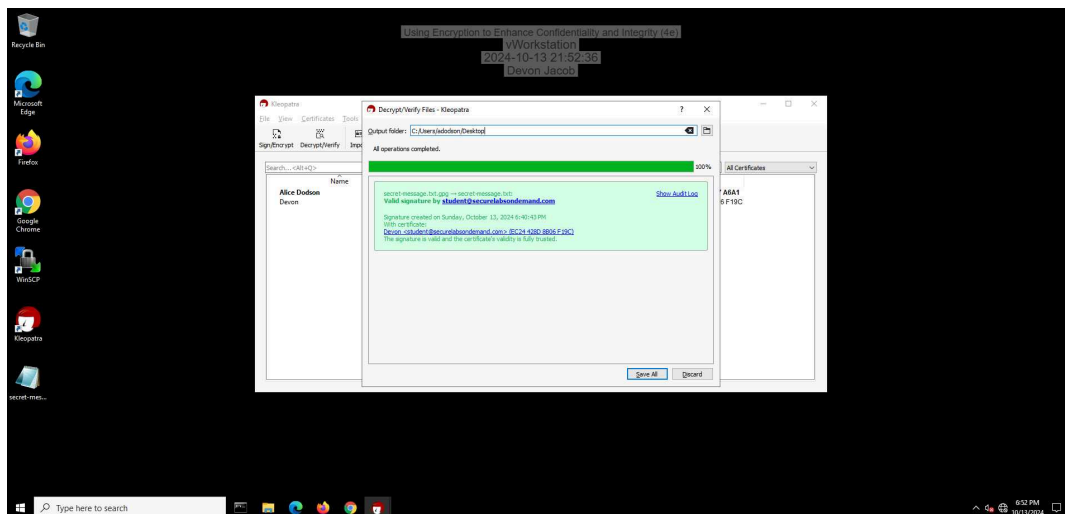
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12. Make a screen capture showing the ciphertext.



Part 3: Decrypt a File Using Asymmetric Encryption

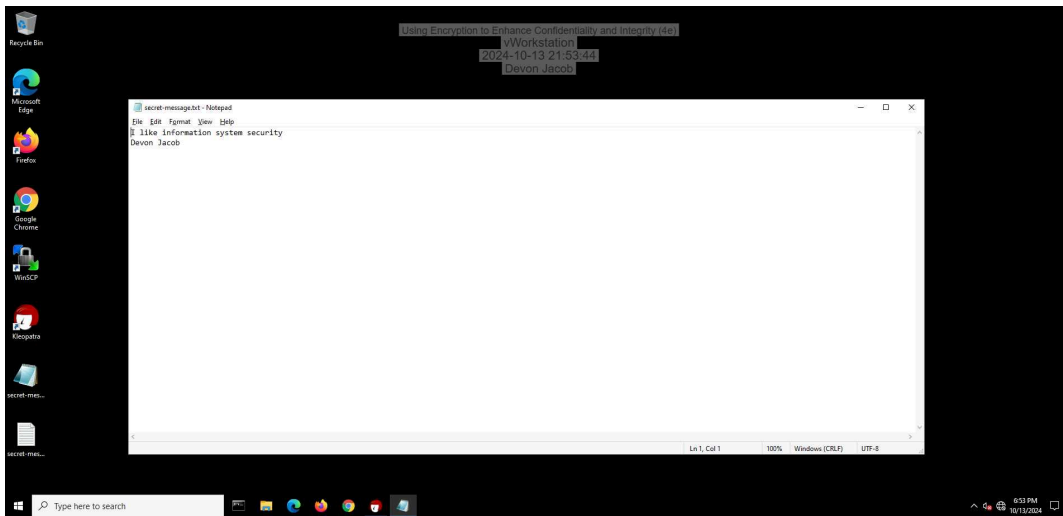
15. Make a screen capture showing the Decrypt/Verify Files window.



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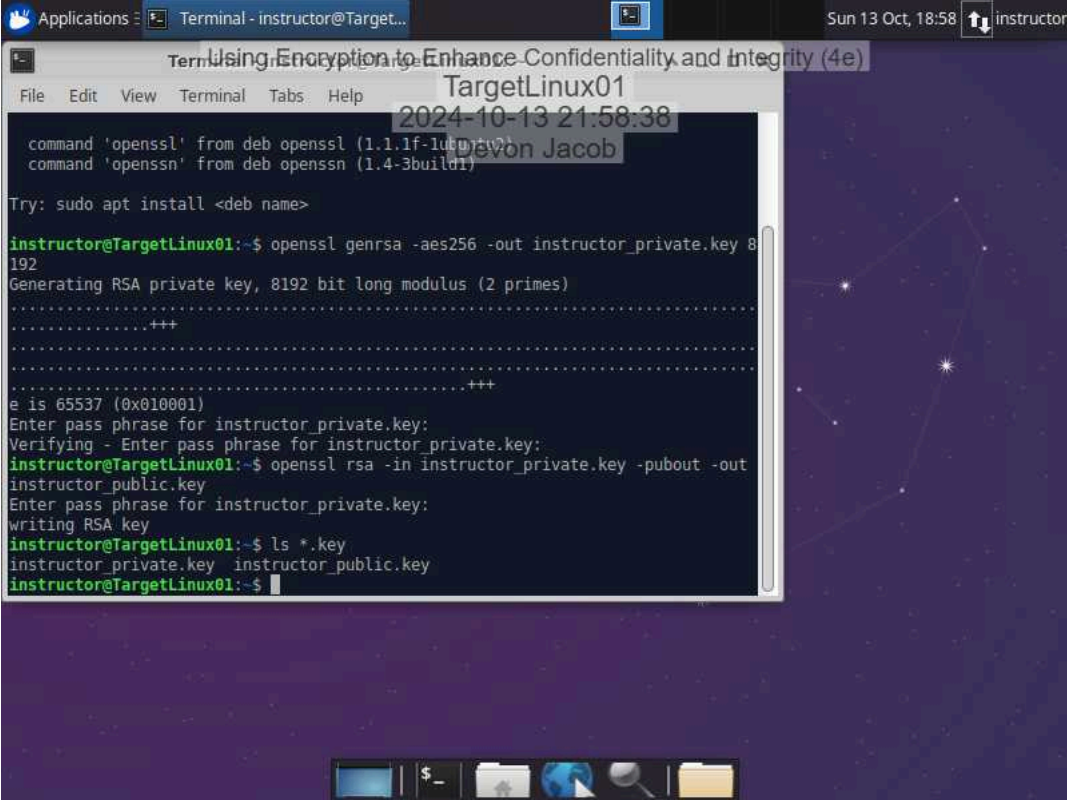
18. Make a screen capture showing the **decrypted secret-message.txt** file in Notepad.



Section 2: Applied Learning

Part 1: Create an Asymmetric Key Pair

10. Make a screen capture showing the instructor's key pair files.



The screenshot shows a terminal window titled 'Terminal - instructor@Target...' with a menu bar (File, Edit, View, Terminal, Tabs, Help). The terminal output shows the installation of OpenSSL and the generation of an asymmetric key pair. The user 'instructor@TargetLinux01' runs the command 'openssl genrsa -aes256 -out instructor_private.key 8192', which generates an 8192-bit RSA private key. The output shows the modulus and the exponent 'e' (65537). The user then runs 'openssl rsa -in instructor_private.key -pubout -out instructor_public.key' to generate the public key. Finally, the user runs 'ls *.key' to list the files, showing 'instructor_private.key' and 'instructor_public.key'. The terminal window is overlaid on a desktop background with a starry night sky and a constellation. The system clock in the top right corner shows 'Sun 13 Oct, 18:58'.

```
command 'openssl' from deb openssl (1.1.1f-1ubuntu1)
command 'openssl' from deb openssl (1.4.3build1)

Try: sudo apt install <deb name>

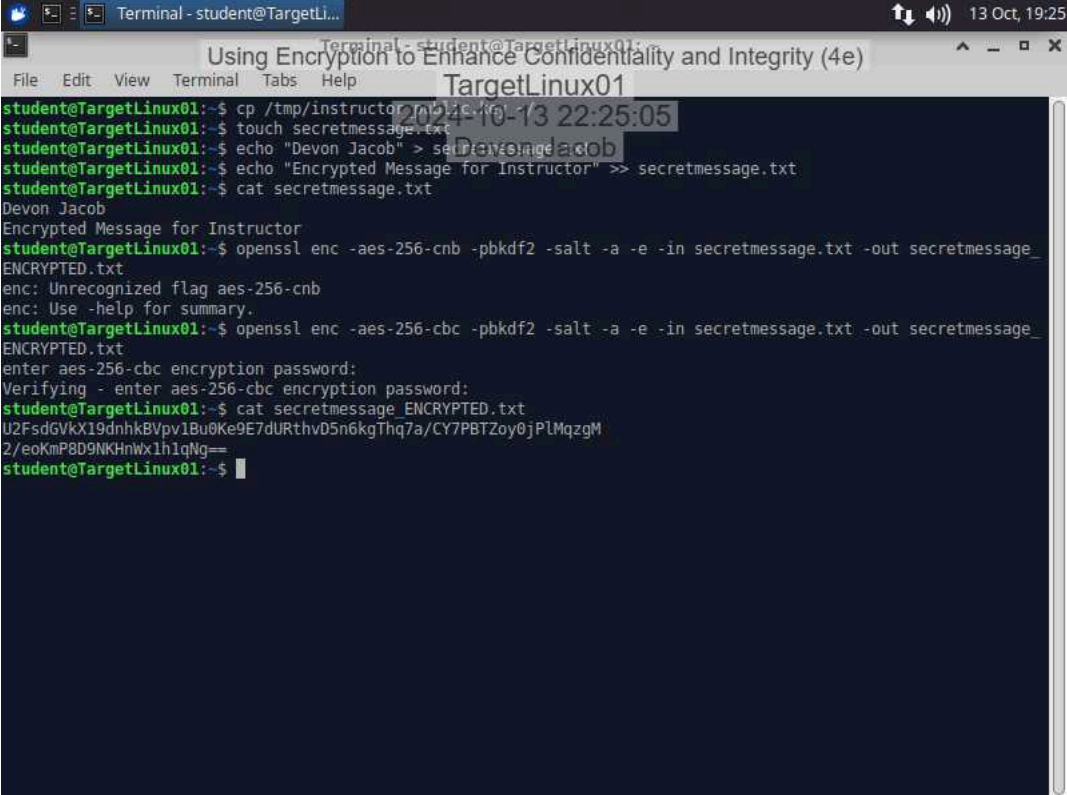
instructor@TargetLinux01:~$ openssl genrsa -aes256 -out instructor_private.key 8192
Generating RSA private key, 8192 bit long modulus (2 primes)
.....+++
.....+++
e is 65537 (0x010001)
Enter pass phrase for instructor_private.key:
Verifying - Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ openssl rsa -in instructor_private.key -pubout -out instructor_public.key
Enter pass phrase for instructor_private.key:
writing RSA key
instructor@TargetLinux01:~$ ls *.key
instructor_private.key  instructor_public.key
instructor@TargetLinux01:~$
```

Part 2: Encrypt a File Using Symmetric Encryption

11. Document the password you used to symmetrically encrypt the file.

yourpassword

13. Make a screen capture showing the ciphertext in the `secretmessage_ENCRYPTED.txt` file.

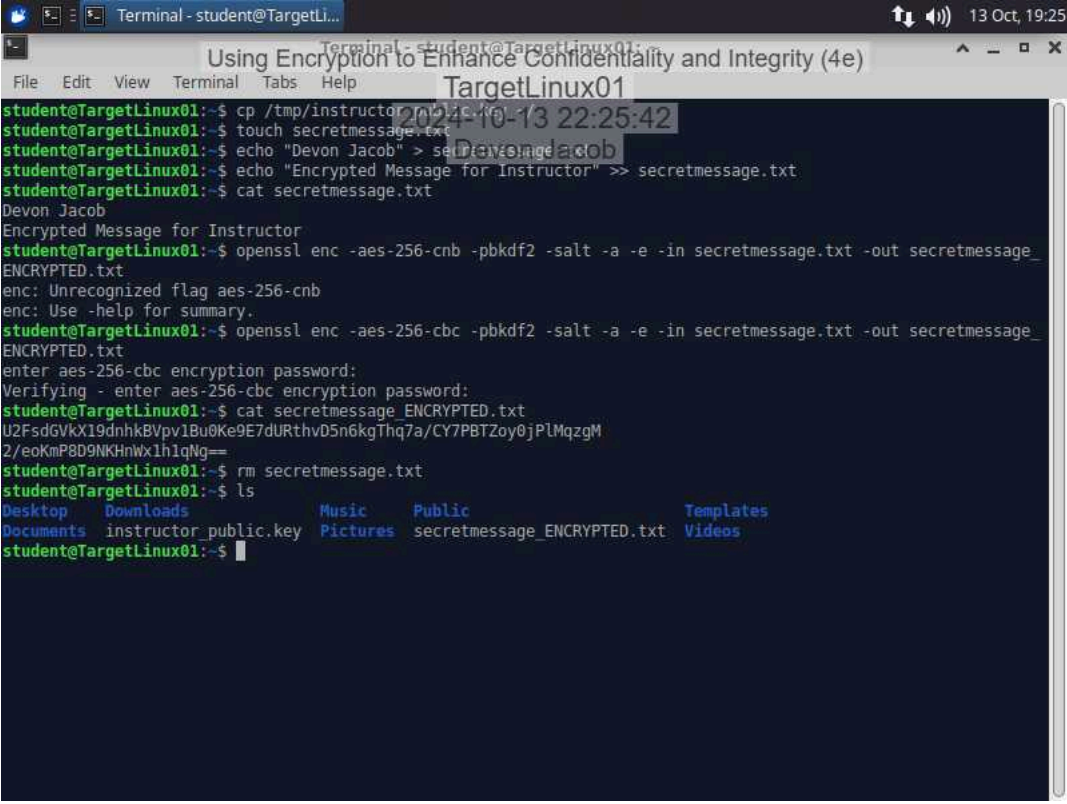
A terminal window titled "Terminal - student@TargetLinux01" with a menu bar (File, Edit, View, Terminal, Tabs, Help) and window controls. The terminal shows a series of commands and their outputs. The user creates a file, adds two lines of text, and then attempts to encrypt it using openssl. The first command uses the flag -cnb, which results in an error. The second command uses the correct flag -cbc, prompts for a password, and successfully creates the encrypted file. Finally, the user cat's the contents of the encrypted file, showing a long base64-encoded string. There are some redaction boxes over the terminal output: a grey box over the timestamp "2024-10-13 22:25:05", a grey box over the name "Devon Jacob", and a grey box over the end of the encrypted output.

```
student@TargetLinux01:~$ cp /tmp/instructor.m24c-10-13-22:25:05
student@TargetLinux01:~$ touch secretmessage.txt
student@TargetLinux01:~$ echo "Devon Jacob" > secretmessage.txt
student@TargetLinux01:~$ echo "Encrypted Message for Instructor" >> secretmessage.txt
student@TargetLinux01:~$ cat secretmessage.txt
Devon Jacob
Encrypted Message for Instructor
student@TargetLinux01:~$ openssl enc -aes-256-cnb -pbkdf2 -salt -a -e -in secretmessage.txt -out secretmessage_ENCRYPTED.txt
enc: Unrecognized flag aes-256-cnb
enc: Use -help for summary.
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.txt -out secretmessage_ENCRYPTED.txt
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.txt
U2FsdGVkX19dnhkBVpv1Bu0Ke9E7dURthvD5n6kgThq7a/CY7PBTZoy0jPlMqzgM
2/eoKmP8D9NKHnWx1hiqNg==
student@TargetLinux01:~$
```

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16. Make a screen capture showing the output of the ls command.

A terminal window titled 'Terminal - student@TargetLinux01' with a menu bar (File, Edit, View, Terminal, Tabs, Help) and a title bar (2024-10-13 22:25:42). The terminal shows a series of commands and their outputs. The user creates a file 'secretmessage.txt', echoes a message into it, and then attempts to encrypt it with 'openssl enc -aes-256-cnb'. This fails with an 'Unrecognized flag' error. The user then successfully encrypts the file using 'openssl enc -aes-256-cbc'. Finally, the user runs 'ls' to show the directory contents, which includes the original file, the encrypted file, and a public key.

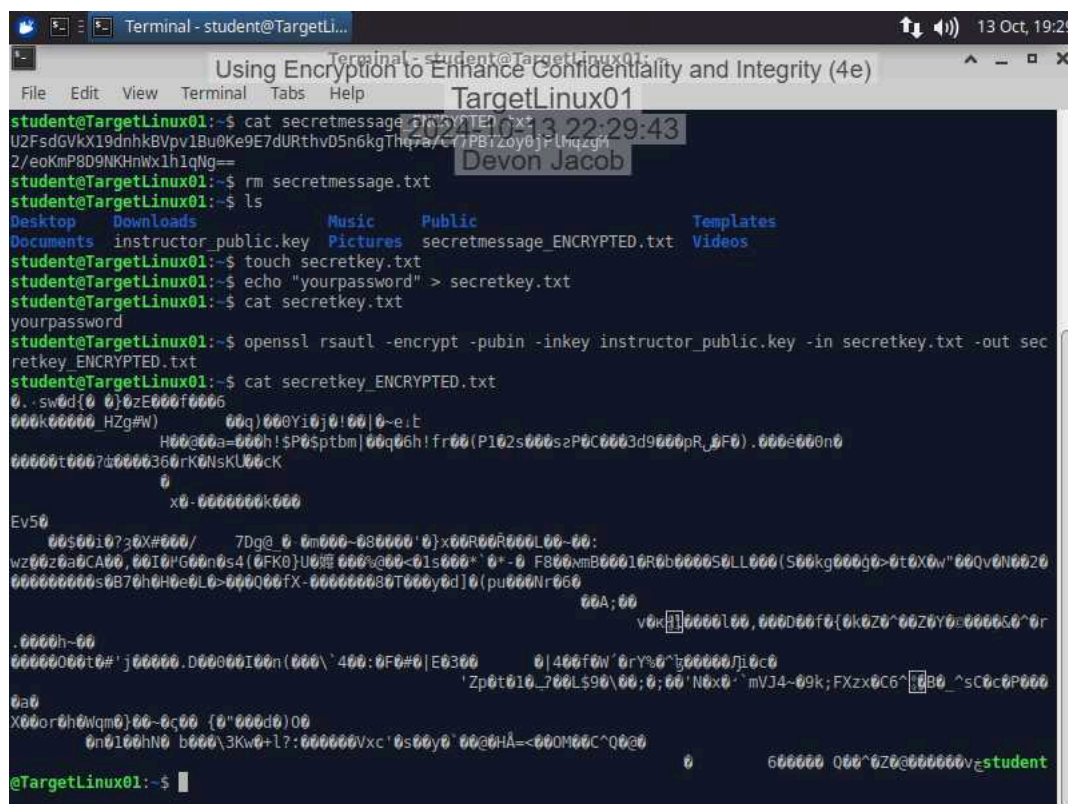
```
student@TargetLinux01:~$ cp /tmp/instructor/public.key .
student@TargetLinux01:~$ touch secretmessage.txt
student@TargetLinux01:~$ echo "Devon Jacob" > secretmessage.txt
student@TargetLinux01:~$ echo "Encrypted Message for Instructor" >> secretmessage.txt
student@TargetLinux01:~$ cat secretmessage.txt
Devon Jacob
Encrypted Message for Instructor
student@TargetLinux01:~$ openssl enc -aes-256-cnb -pbkdf2 -salt -a -e -in secretmessage.txt -out secretmessage_
ENCRIPTED.txt
enc: Unrecognized flag aes-256-cnb
enc: Use -help for summary.
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.txt -out secretmessage_
ENCRIPTED.txt
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.txt
U2FsdGVkX19dnhkBVpv1Bu0Ke9E7dURthvD5n6kgThq7a/CY7PBTZoy0jPLMqzgM
Z/eoKmP8D9NKHnWx1h1qNg==
student@TargetLinux01:~$ rm secretmessage.txt
student@TargetLinux01:~$ ls
Desktop  Downloads  Music      Public      Templates
Documents instructor_public.key Pictures  secretmessage_ENCRYPTED.txt Videos
student@TargetLinux01:~$
```

Part 3: Transfer and Decrypt a File Using Hybrid Cryptography

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6. Make a screen capture showing the encrypted contents of the `secretkey_ENCRYPTED.txt` file.

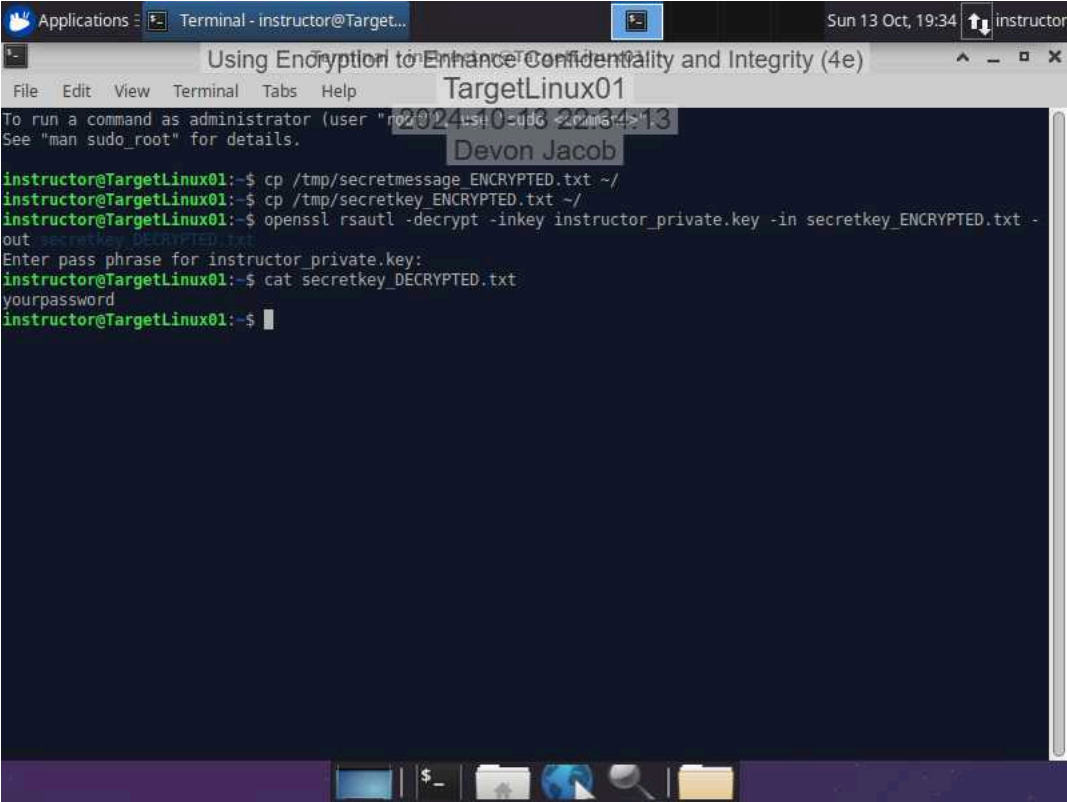


```
Terminal - student@TargetLinux01
Using Encryption to Enhance Confidentiality and Integrity (4e)
TargetLinux01
2024-10-13 22:29:43
Devon Jacob
student@TargetLinux01:~$ cat secretmessage.txt
U2FsdGVkX19dnhkBVpvl8u0Ke9E7dURthvD5n6kgThera/C7P8Tzoy0jFUnqzgn
2/eoKmP8D9NKHnWx1h1qNg==
student@TargetLinux01:~$ rm secretmessage.txt
student@TargetLinux01:~$ ls
Desktop  Downloads  Music  Public  Templates
Documents  instructor_public.key  Pictures  secretmessage_ENCRYPTED.txt  Videos
student@TargetLinux01:~$ touch secretkey.txt
student@TargetLinux01:~$ echo "yourpassword" > secretkey.txt
student@TargetLinux01:~$ cat secretkey.txt
yourpassword
student@TargetLinux01:~$ openssl rsautl -encrypt -pubin -inkey instructor_public.key -in secretkey.txt -out secretkey_ENCRYPTED.txt
student@TargetLinux01:~$ cat secretkey_ENCRYPTED.txt
0..sw0d{0 0}0ZE000f0006
000k00000_HZg#W) 00q)000Yi0!00|0~e:t
H00@00a=000h!$P0$ptbm|00q06h!fr00(P102s000saP0C0003d9000pR_0F0).000e000n0
00000t0007d0000360rk0NsKL00ck
0
x0-0000000k000
Ev50
00$00i0?0X#000/ 7Dg@ 0 0m000-00000'0}x00R00R000L00-00:
wz00z0a0CA00,00I0"G00n0s4(0FK0}U0牌 000%00<01s000*"0+-0 F800mb00010R0b0000S0LL000(S00kg000g0>0t0X0w"00Qv0N0020
00000000s0B70h0H0e0L0>000Q00fX-000000080T000y0d]0(pu000Nr060
00A;00
v0k0000l00,000D00f0{0k0Z0^00Z0Y0:0000S0^0r
.0000h~00
00000000t0# j00000.D0000I00n(000\`400:0F0#0[E0300 0|400f0W'0rY%0^b00000ji0c0
'Zp0t0l0_700L$90/00;0;00'N0x0`mVJ4~09k;FXzx0C6^0B0_^sC0c0P000
0a0
X000r0h0Wqm0}00-0c00 {0"000d0)00
0n0l00hN0 b000\3Kw0+l?:000000Vxc'0s00y0 00@0HA=<000M00C^Q0@0
0 600000 000^0Z0@000000v0student
@TargetLinux01:~$
```

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17. Make a screen capture showing the **decrypted contents of the secretkey_DECRYPTED.txt file.**

A terminal window titled 'Terminal - instructor@Target...' is shown. The window has a menu bar with 'File', 'Edit', 'View', 'Terminal', 'Tabs', and 'Help'. The title bar also includes 'Using Encryption to Enhance Confidentiality and Integrity (4e)', 'TargetLinux01', and a timestamp '2024-10-13 22:34:13'. The terminal content shows the following commands and output:

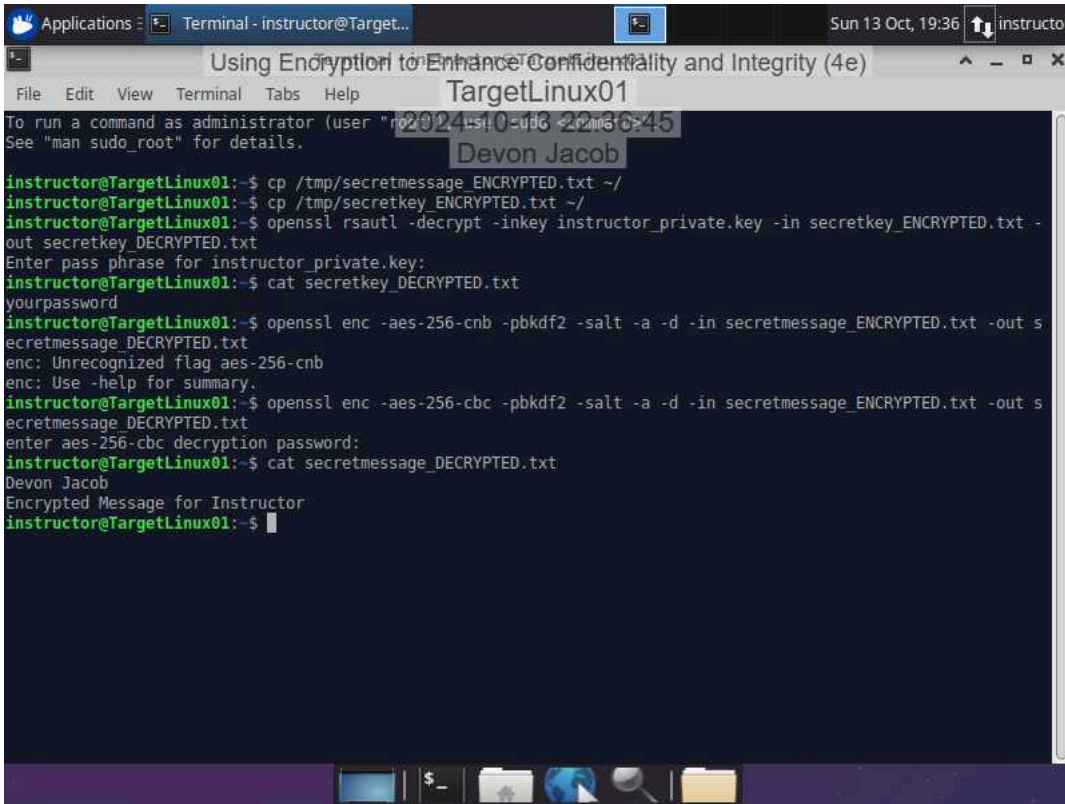
```
instructor@TargetLinux01:~$ cp /tmp/secretmessage ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ cp /tmp/secretkey ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.key -in secretkey_ENCRYPTED.txt -
out secretkey_DECRYPTED.txt
Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ cat secretkey_DECRYPTED.txt
yourpassword
instructor@TargetLinux01:~$
```

The terminal window is overlaid on a desktop environment with a taskbar at the bottom containing icons for a terminal, file manager, and other applications. The desktop background is dark blue.

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21. Make a screen capture showing the contents of the `secretmessage_DECRYPTED` file.



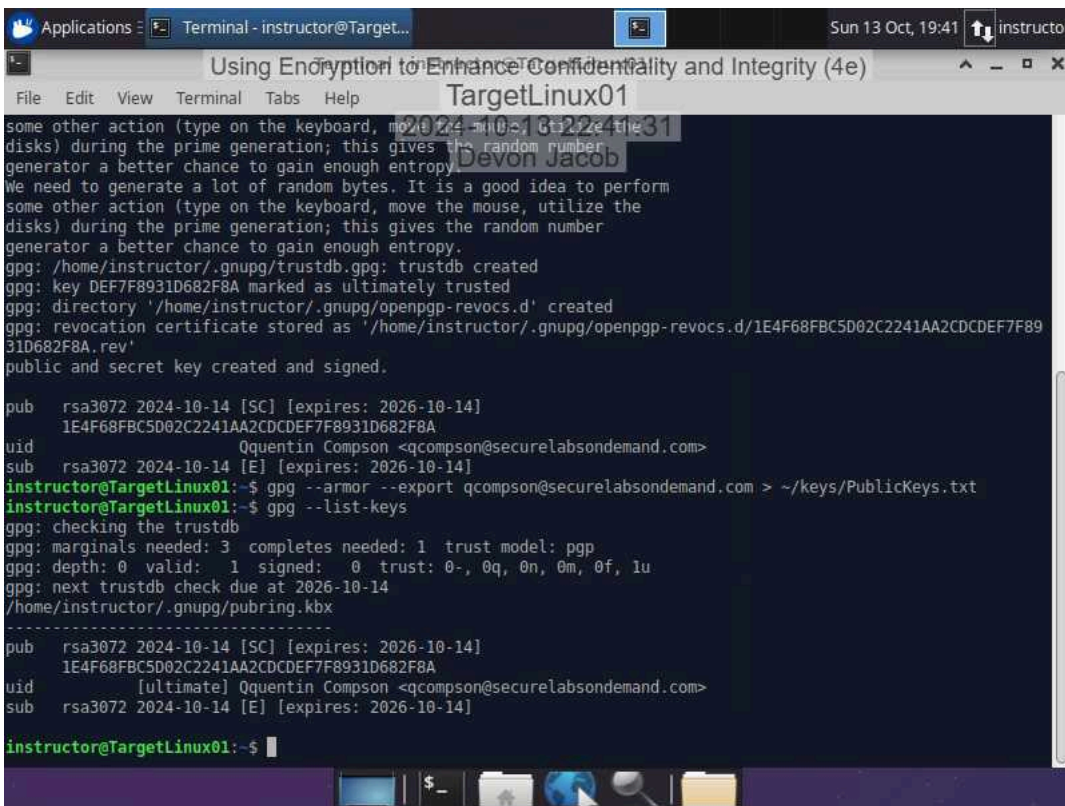
The image shows a terminal window titled "Terminal - instructor@Target..." with a date and time of "Sun 13 Oct, 19:36". The window contains the following commands and output:

```
instructor@TargetLinux01:~$ cp /tmp/secretmessage_ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ cp /tmp/secretkey_ENCRYPTED.txt ~/
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.key -in secretkey_ENCRYPTED.txt -
out secretkey_DECRYPTED.txt
Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ cat secretkey_DECRYPTED.txt
yourpassword
instructor@TargetLinux01:~$ openssl enc -aes-256-cnb -pbkdf2 -salt -a -d -in secretmessage_ENCRYPTED.txt -out s
ecretmessage_DECRYPTED.txt
enc: Unrecognized flag aes-256-cnb
enc: Use -help for summary.
instructor@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -d -in secretmessage_ENCRYPTED.txt -out s
ecretmessage_DECRYPTED.txt
enter aes-256-cbc decryption password:
instructor@TargetLinux01:~$ cat secretmessage_DECRYPTED.txt
Devon Jacob
Encrypted Message for Instructor
instructor@TargetLinux01:~$
```

Section 3: Challenge and Analysis

Part 1: Digitally Sign a Document Using GPG

Make a screen capture showing the **key fingerprint** for the key pair you generated in this part of the lab.



```
Applications ▢ Terminal - instructor@TargetLinux01 Sun 13 Oct, 19:41 instructor
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TargetLinux01
some other action (type on the keyboard, move the mouse, etc.) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/instructor/.gnupg/trustdb.gpg: trustdb created
gpg: key DEF7F8931D682F8A marked as ultimately trusted
gpg: directory '/home/instructor/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/instructor/.gnupg/openpgp-revocs.d/1E4F68FBC5D02C2241AA2CDCDEF7F89
31D682F8A.rev'
public and secret key created and signed.

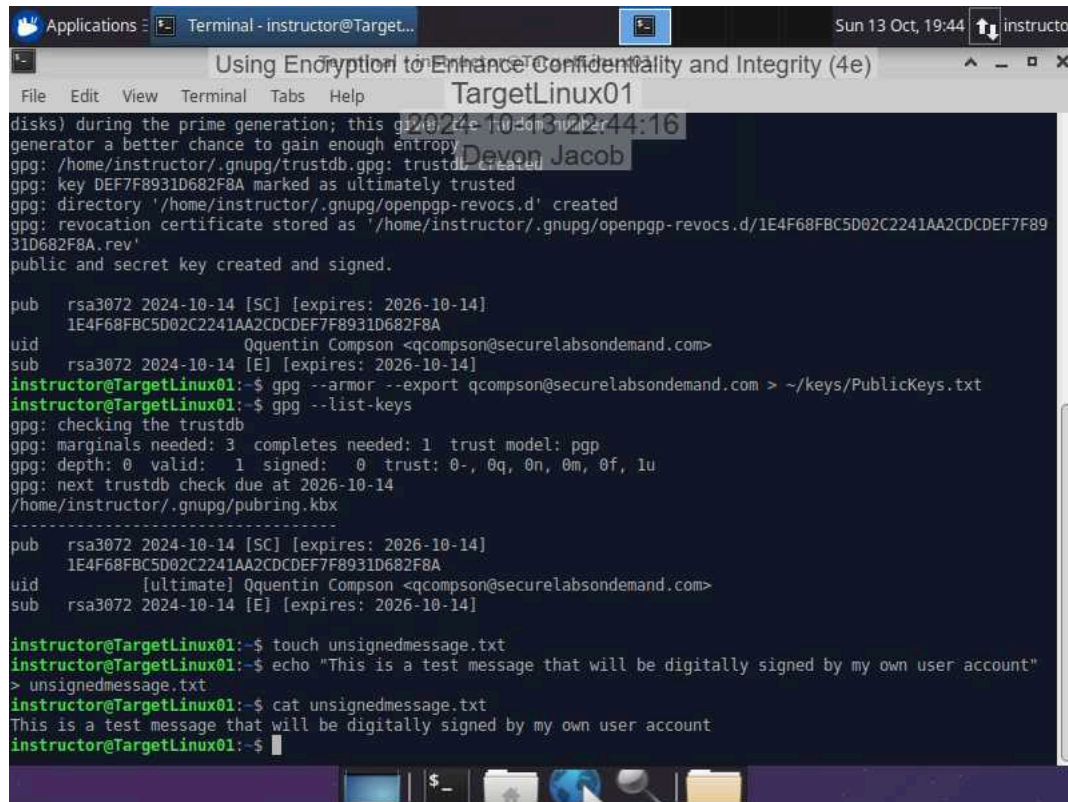
pub  rsa3072 2024-10-14 [SC] [expires: 2026-10-14]
     1E4F68FBC5D02C2241AA2CDCDEF7F8931D682F8A
uid                               Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2024-10-14 [E] [expires: 2026-10-14]
instructor@TargetLinux01:~$ gpg --armor --export qcompson@securelabsondemand.com > ~/keys/PublicKeys.txt
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2026-10-14
/home/instructor/.gnupg/pubring.kbx
-----
pub  rsa3072 2024-10-14 [SC] [expires: 2026-10-14]
     1E4F68FBC5D02C2241AA2CDCDEF7F8931D682F8A
uid                               [ultimate] Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2024-10-14 [E] [expires: 2026-10-14]

instructor@TargetLinux01:~$
```

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Make a screen capture showing the contents of the unsignedmessage.txt file.



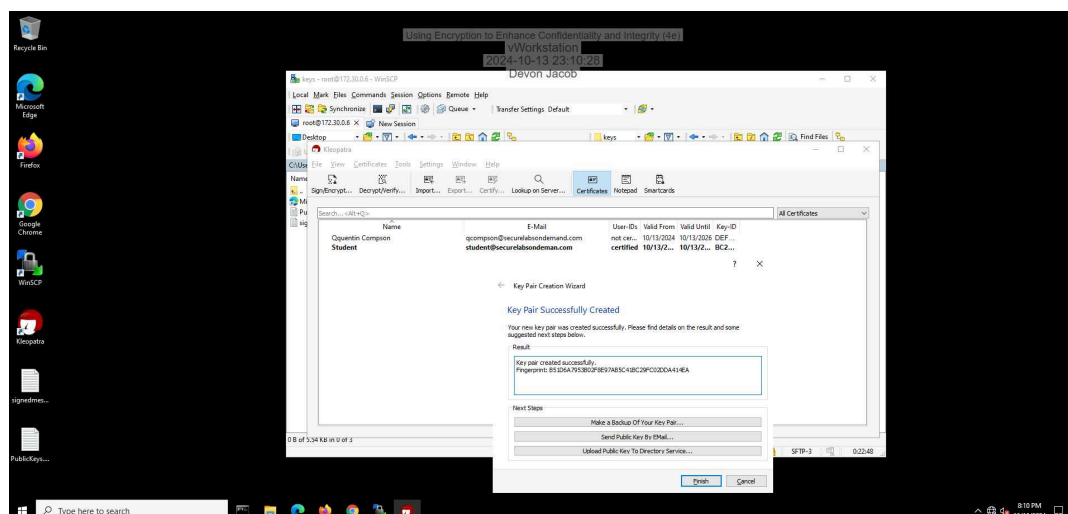
```
disks) during the prime generation; this gives the
generator a better chance to gain enough entropy.
gpg: /home/instructor/.gnupg/trustdb.gpg: trust database created
gpg: key DEF7F8931D682F8A marked as ultimately trusted
gpg: directory '/home/instructor/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/instructor/.gnupg/openpgp-revocs.d/1E4F68FBC5D02C2241AA2CDCE7F8931D682F8A.rev'
public and secret key created and signed.

pub  rsa3072 2024-10-14 [SC] [expires: 2026-10-14]
     1E4F68FBC5D02C2241AA2CDCE7F8931D682F8A
uid                               Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2024-10-14 [E] [expires: 2026-10-14]
instructor@TargetLinux01:~$ gpg --armor --export qcompson@securelabsondemand.com > ~/keys/PublicKeys.txt
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2026-10-14
/home/instructor/.gnupg/pubring.kbx
-----
pub  rsa3072 2024-10-14 [SC] [expires: 2026-10-14]
     1E4F68FBC5D02C2241AA2CDCE7F8931D682F8A
uid                               [ultimate] Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2024-10-14 [E] [expires: 2026-10-14]

instructor@TargetLinux01:~$ touch unsignedmessage.txt
instructor@TargetLinux01:~$ echo "This is a test message that will be digitally signed by my own user account"
> unsignedmessage.txt
instructor@TargetLinux01:~$ cat unsignedmessage.txt
This is a test message that will be digitally signed by my own user account
instructor@TargetLinux01:~$
```

Part 2: Verify the Digital Signature Using Kleopatra

Make a screen capture showing the successful signature verification on the signed message file.



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